lander by USA, but the onboard systems malfunctioned and the probe did not beam back any scientific data. The Chang'e 4 mission is the first spacecraft to execute a soft landing on the Lunar far side, which means a controlled descent.

The Chang'e 4 probe consisted of a lander and a rover. The landing site was the Sinus Iridum or the Bay of Rainbows, within the Von Karman crater, which itself sits in a bigger impact crater called the South Pole-Aitken Basin. At a height of 2 kilometers, the probe halted its descent to identify any large obstacles, such as craters and rocks. At a height of about a 100 meters above the surface, the hover momentarily hovered to identify smaller obstacles such as rocks, and measure the slopes on the surface. The probe then autonomously selected a relatively smooth surface for landing. A camera on board the spacecraft took photos during the landing, which lasted 12 minutes. The landing was executed without any control from Earth, but as soon as the probe touched down on the surface, the solar panels and the antennas were unfolded, through instructions relayed through Queqiao. Soon after touchdown, the first image from the surface of the far side of the Moon was beamed back to the Earth, showing the direction in which the rover would explore.

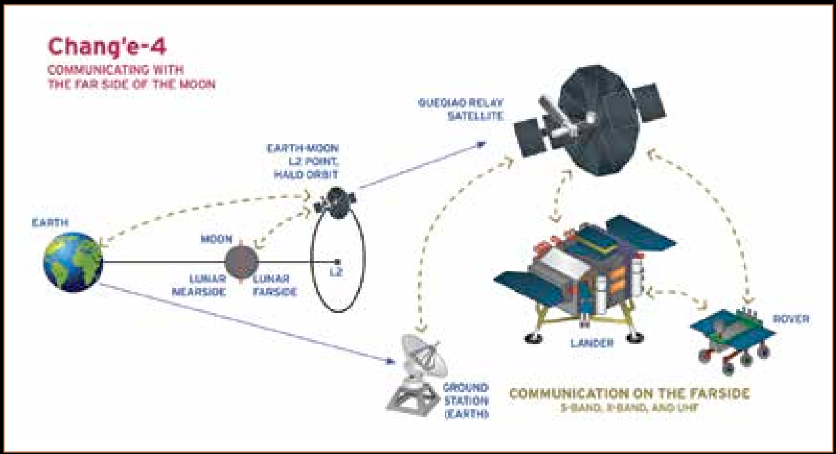
Apart from the cameras to image the surface, there are three scientific payloads on the lander. A low frequency spectrometer will study the Lunar ionosphere, and examine solar radio bursts. A neutron dosimeter developed in Germany will study the radiation dosage of the Moon, for the benefit of future manned missions. Perhaps the most interesting experiment on board the lander is the Lunar Micro Ecosystem, a sealed biosphere carrying insect eggs and seeds. On board were four types of plants, seeds and fruit fly eggs. The biosphere was designed to be self contained. The experiment was planned to run for 100 days, but was prematurely terminated because the biosphere got too cold. However, cotton and potato seeds sprouted, and important information was obtained.

THE YUTU-2 ROVER

The Yutu-2 Rover, which translates to white jade rabbit in English, is named after the pet of the Chinese Moon goddess. On board the rover is a panoramic camera that is mounted on a mast that can rotate 360 degrees. The camera itself captures stereoscopic images. There is a ground penetrating radar for probing the surface, and a Visible and Near-Infrared Imaging Spectrometer (VNIS) for studying the tenuous atmosphere of the Moon.

Both the probe and the rover are equipped with radioisotope heater units (RHUs), that uses the decay of plutonium-238 to keep the systems warm during the long nights on the Moon, when temperatures can drop to -183 degrees Celsius. A neutral atom analyser will measure how particles in the solar wind interact with the Moon, and perhaps find clues on the origin of water on the surface. The probe will study the rock formations and chemical composition of the Lunar surface.

The rover on the surface



How Queqiao Operates

After the successful landing, Sun Zezhou, chief designer of the Chang'e-4 mission said, "If we want to build a scientific research station on the Moon, we will need to land multiple probes within the same area so that they can be assembled easily into a complex, which requires even greater landing accuracy. So solving the challenges of the Chang'e-4 mission can lay the foundation for the following lunar exploration and future landing on other planets."

The Chang'e 4 is a part of the Chinese Lunar Exploration Program, which is a series of increasingly ambitious missions for exploring the moon. The Chang'e 1 and 2 were orbiters that surveyed the Moon for follow up missions. The Chang'e 3 and 4 had rovers that explored the near and far side of the moon. Finally the Chang'e 5 and 6 are sample return missions that scheduled for launch in 2019. These missions aim to bring back about 4 kilograms of Lunar material, for further study on the Earth. The Chang'e program is a precursor to a Chinese manned mission to the Moon, planned for the early 2030s. d